HISTORICAL PERFORMANCE EXPLORER

Olympic Games: who wins, and how efficiently

30 nations · 17 Games · 1992-2022



Taiwo Ajayi
Report author

Source: Project_06_Olympic_Games.xlsx (as loaded in the model) · 26,824 entries, 1,036 medals

Start the story

The brief: find what separates medal-winning nations from the rest

Aim
Help Olympic committees and sport funders see which nations win, how efficiently they turn entries into medals, and which factors go with success.

Questions this report answers

- How concentrated are medals among nations?
- Which nations convert entries into medals best?
- Do population, wealth and region explain success?
- Has the medal haul changed over time, and do hosts win more at home?
- Which sports and athletes win medals?

30
nations

17
Games (8 Summer, 9 Winter)

1,800
athletes

26,824
entries

20 sports and 139 events, Barcelona 1992 to Beijing 2022.
The source workbook is missing; numbers come from the data stored in the model.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 2

1,036 medals from 26,824 entries, and nine nations win 61% of them

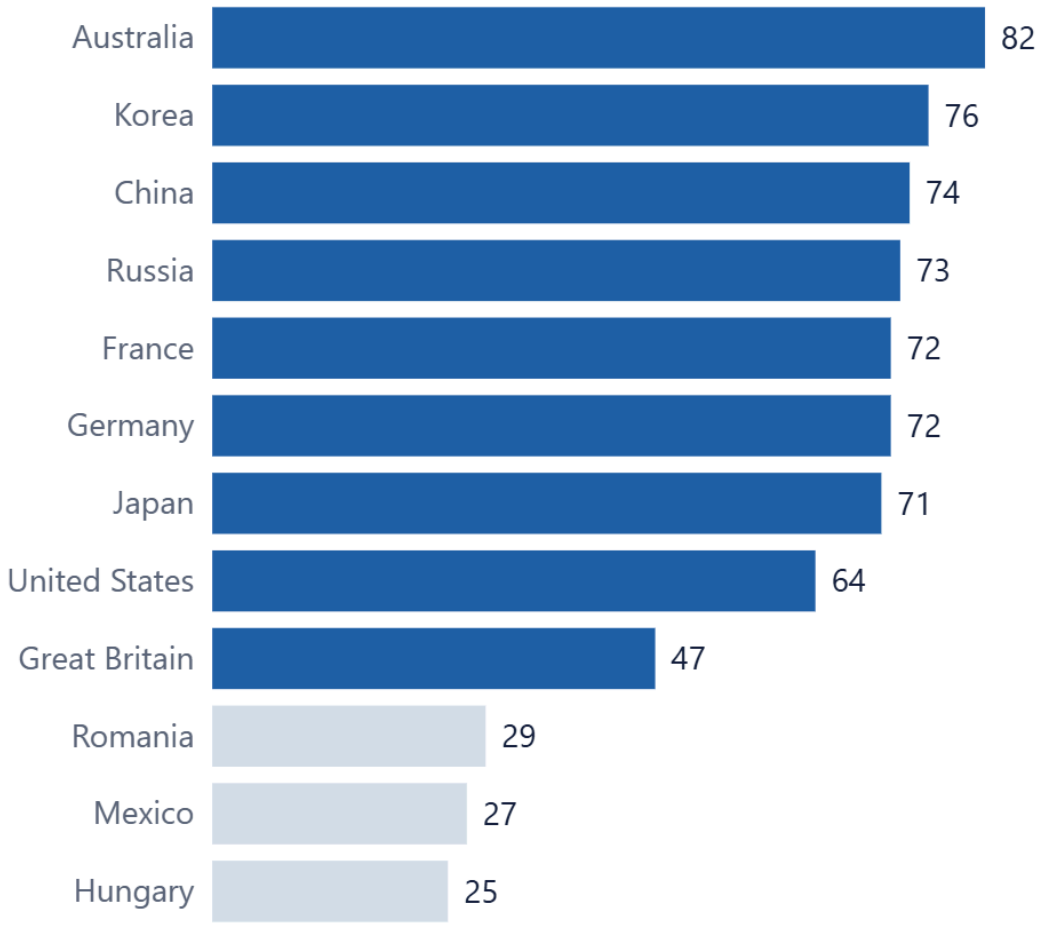
1,036	334	26,824	3.9%
medals won	gold medals	athlete entries	of entries win a medal

CONCENTRATED	**Nine nations**	Win 631 of the 1,036 medals (61%) and 62% of the golds
EFFICIENT	**Conversion**	The leaders turn 6-9% of entries into medals; everyone else 1-3%
SURPRISE	**Small nations**	Jamaica wins 6.2 medals per million people, New Zealand 4.3
NO EFFECT	**Home advantage**	Hosts won 2.75 medals at home against 3.15 in an average Games
BALANCED	**Women and men**	Women won 492 medals, men 546, from equal athlete numbers

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 3

Nine nations win 61% of all medals; Australia leads with 82

Medals by nation, top 12



Medal table, top 10

Nation	Gold	Silver	Bronze	Total	Share
Australia	26	31	25	82	7.9%
Korea	20	26	30	76	7.3%
China	23	18	33	74	7.1%
Russia	25	23	25	73	7.0%
France	24	22	26	72	6.9%
Germany	23	20	29	72	6.9%
Japan	27	23	21	71	6.9%
United States	24	20	20	64	6.2%
Great Britain	15	12	20	47	4.5%
Romania	5	10	14	29	2.8%

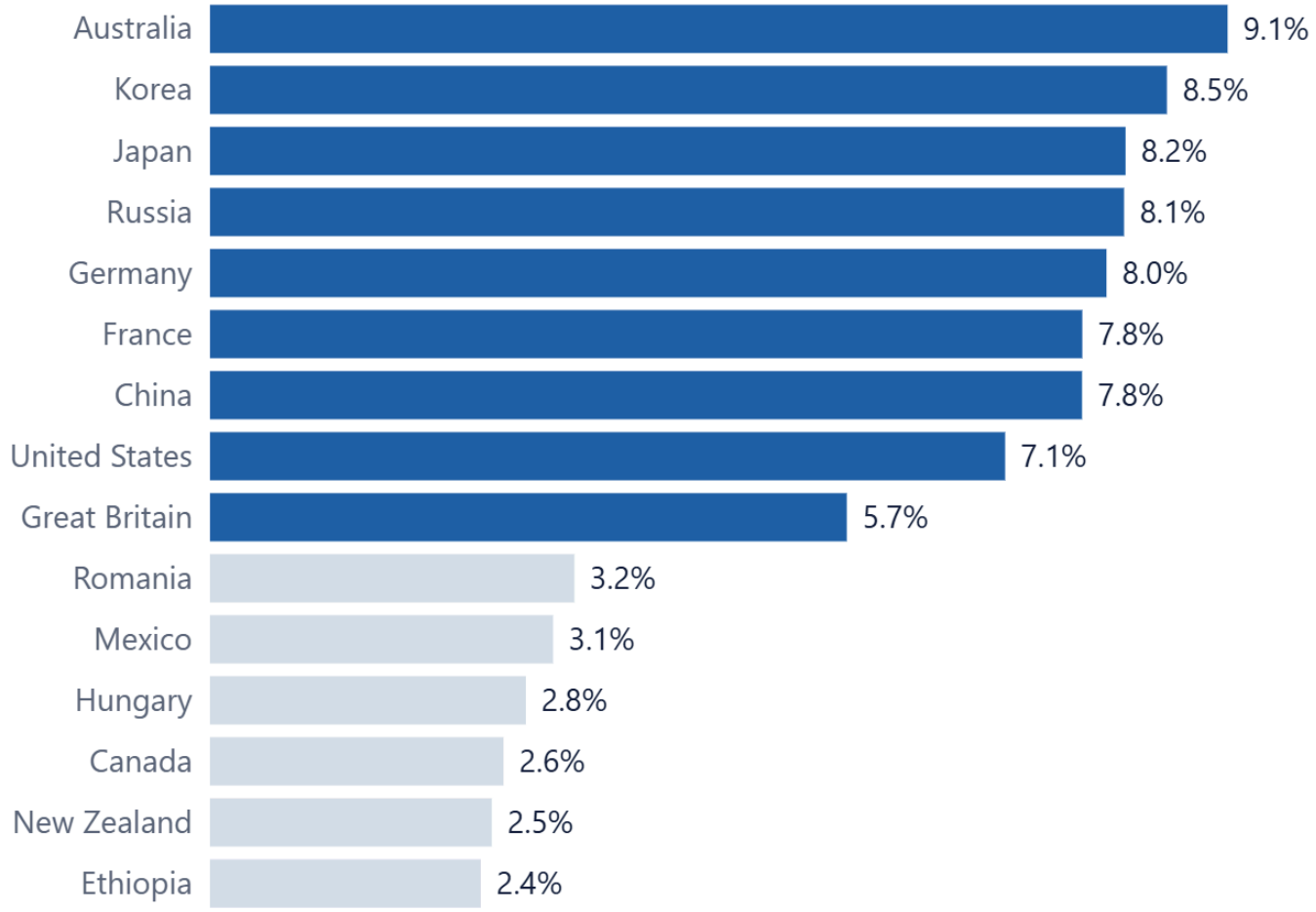
A clear gap after ninth place.

The nine leaders win 47-82 medals and 207 of the 334 golds; the next nation, Romania, wins 29.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 4

The leaders turn 6-9% of their entries into medals; every other nation 1-3%

Medals per 100 entries, by nation (top 15)



26,824
entries, about 900 per nation

3.9%
conversion, all nations

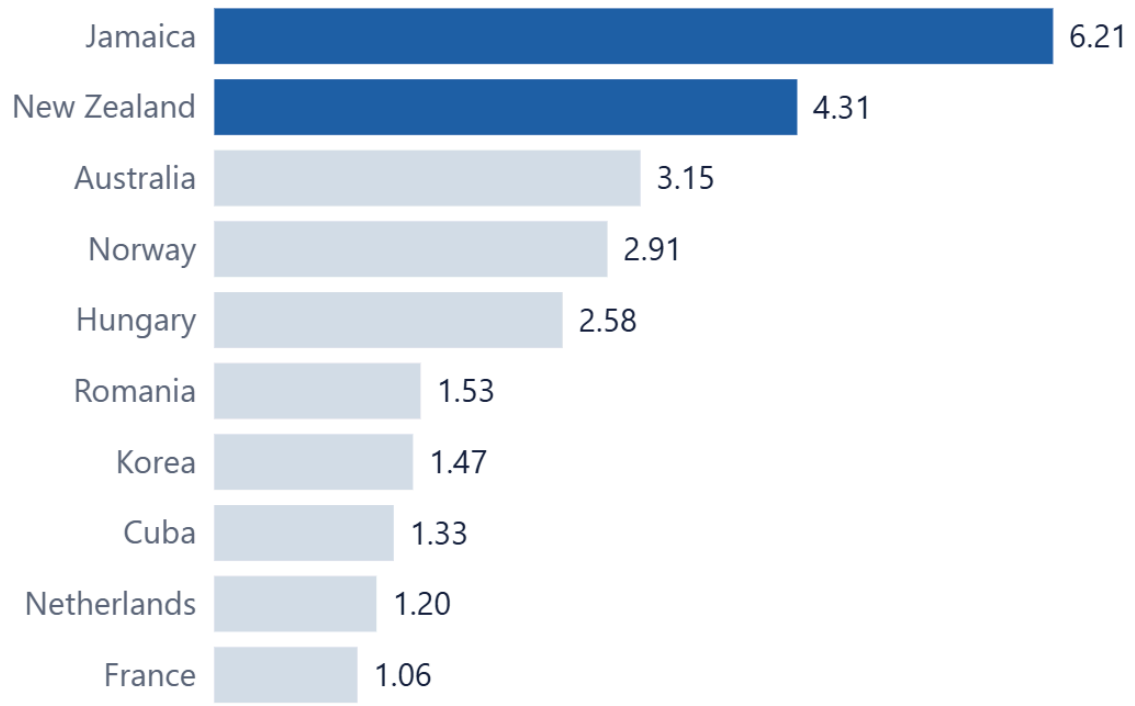
Every nation sends about the same number of entries.
The gap is in what those entries achieve, not how many there are.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 5

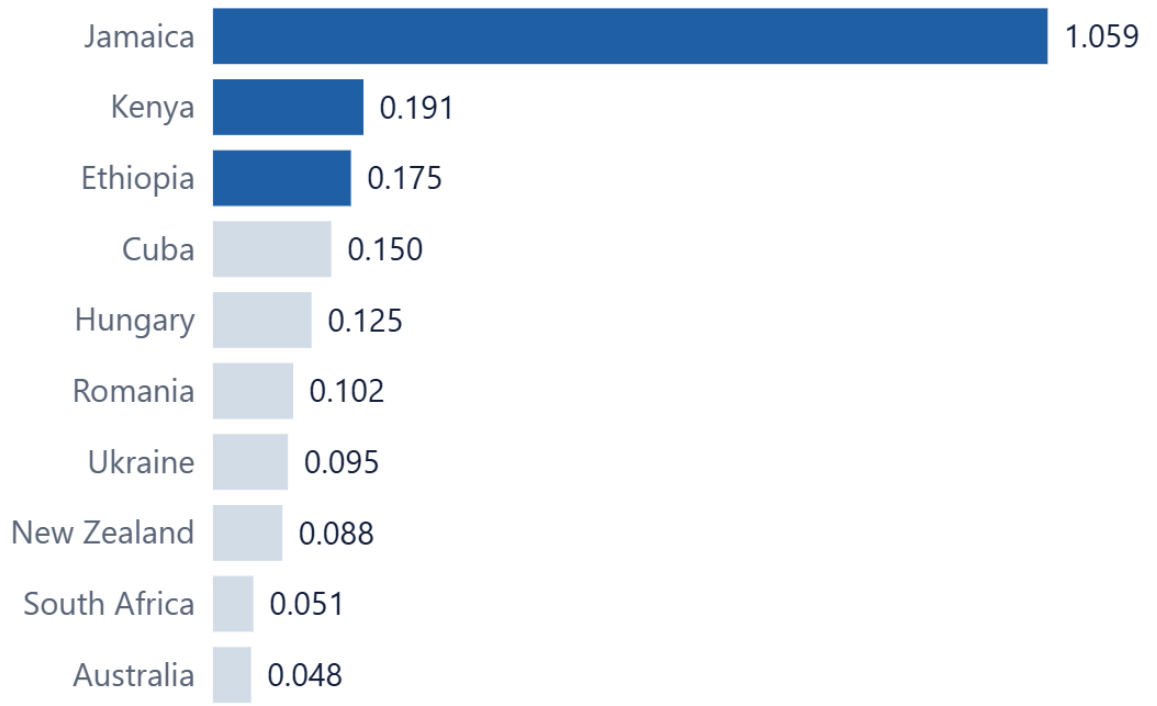
Dashboards

Small nations punch above their weight: Jamaica wins 6.2 medals per million people

Medals per million people, top 10



Medals per $bn of GDP, top 10

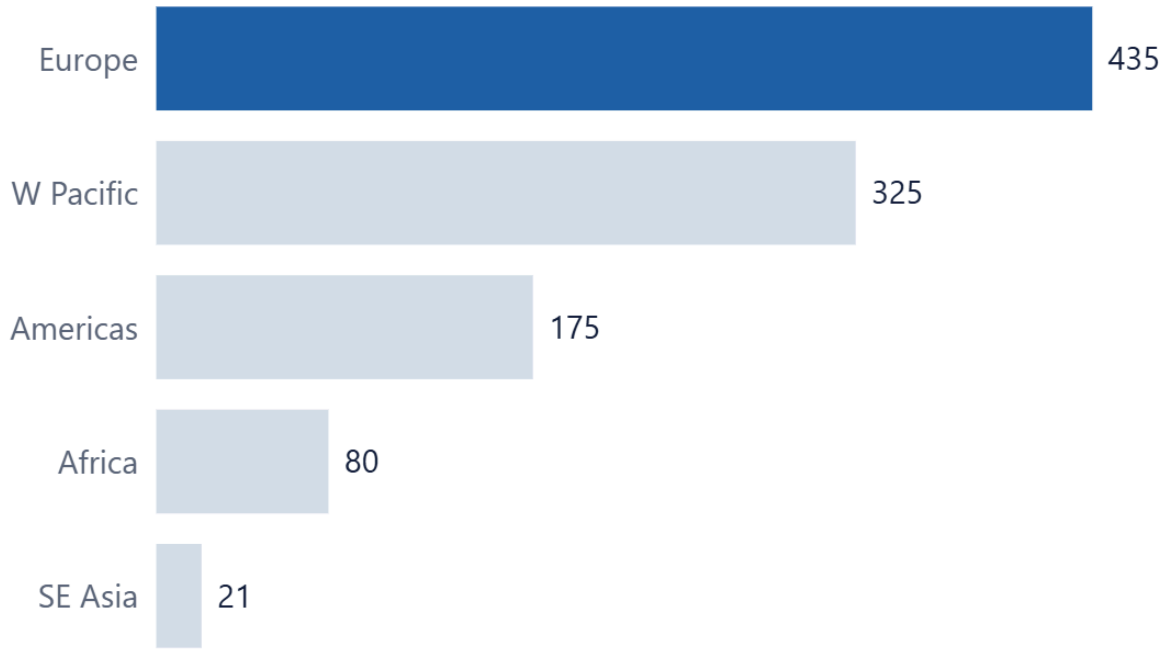


China and India sit near the bottom on both measures: size alone does not win medals.

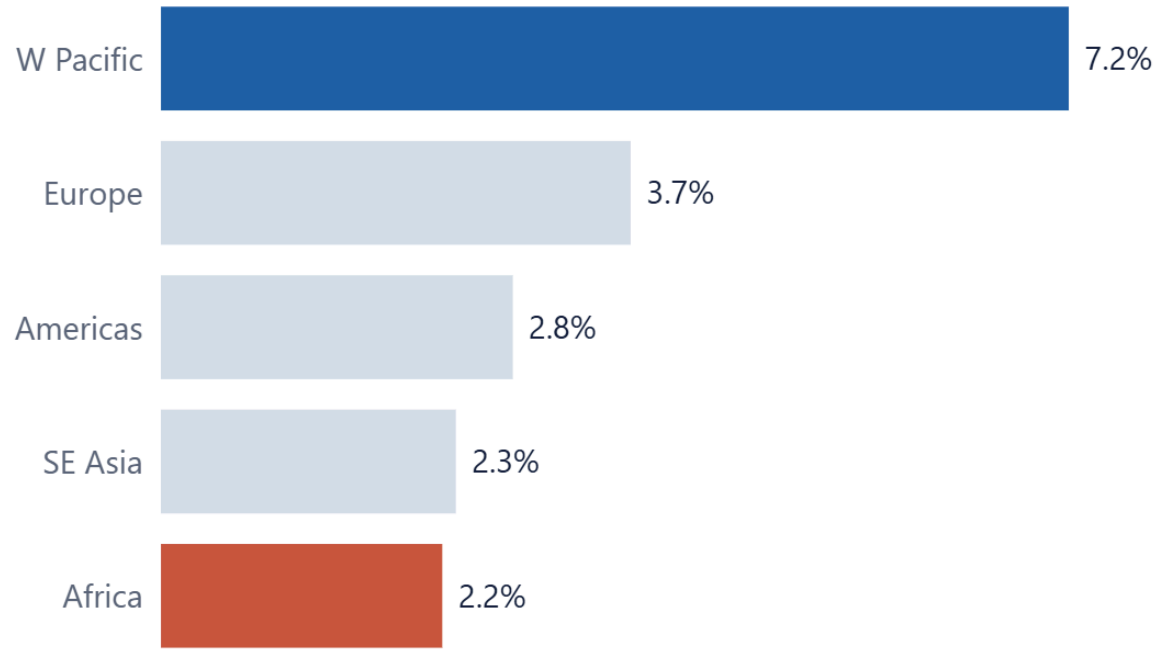
Source: Olympics model (Project_06_Olympic_Games.xlsx) · population and GDP at the latest Games · 6

W Pacific nations convert 7.2% of entries into medals; African nations 2.2%

Medals by region



Medals per 100 entries, by region

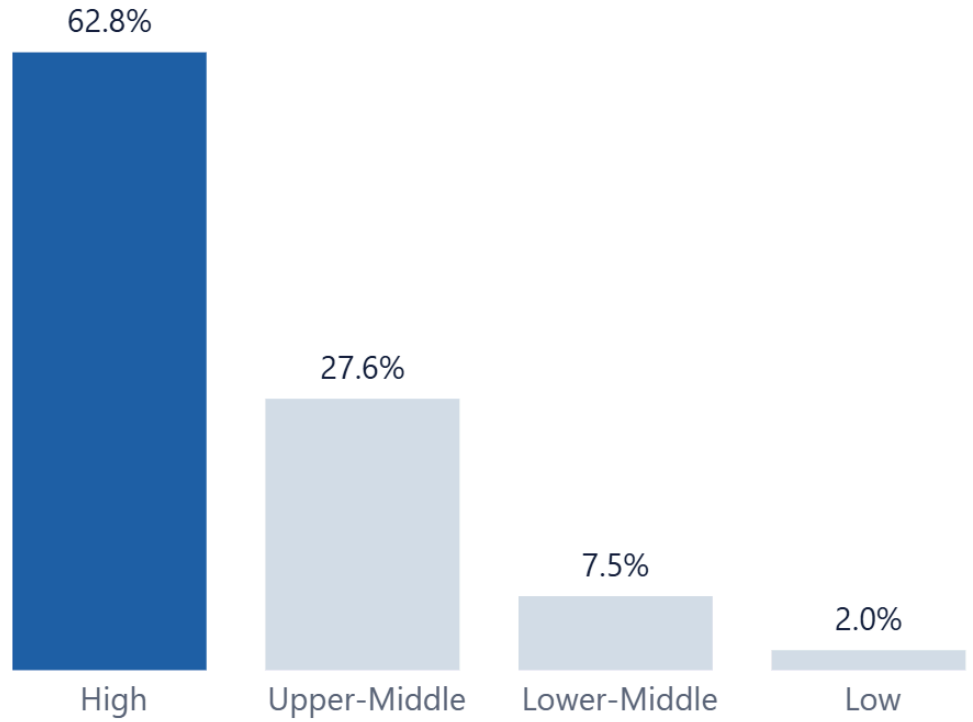


Europe wins the most because it sends 13 nations; W Pacific wins more per entry than anyone.

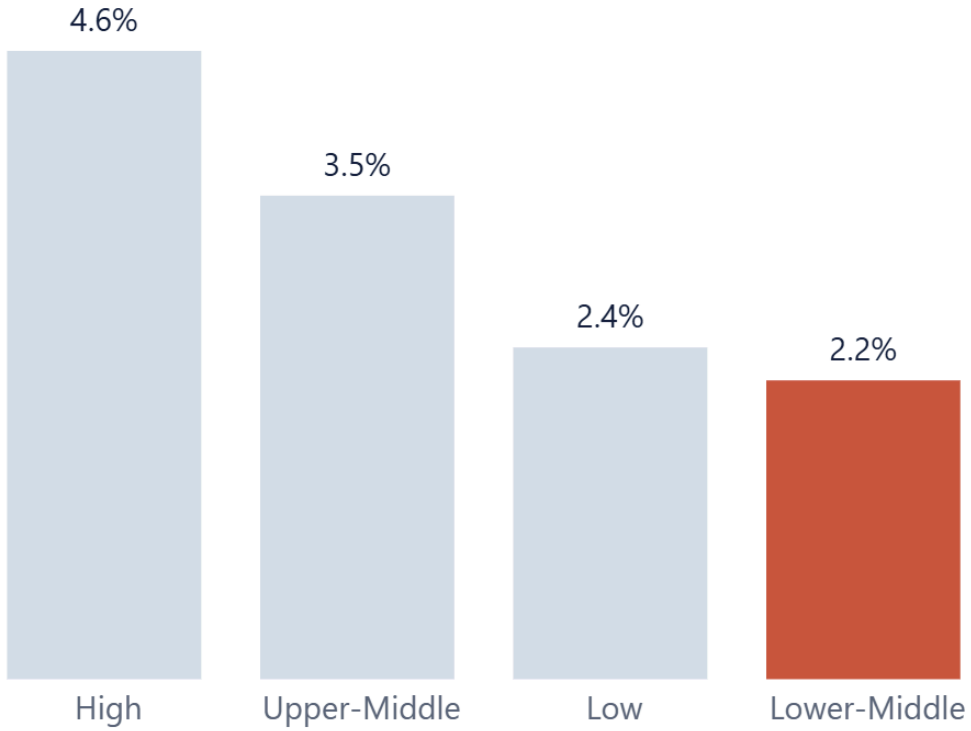
Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 7

High-income nations win 63% of medals and convert twice as well as poorer ones

Share of medals by income group



Medals per 100 entries, by income group

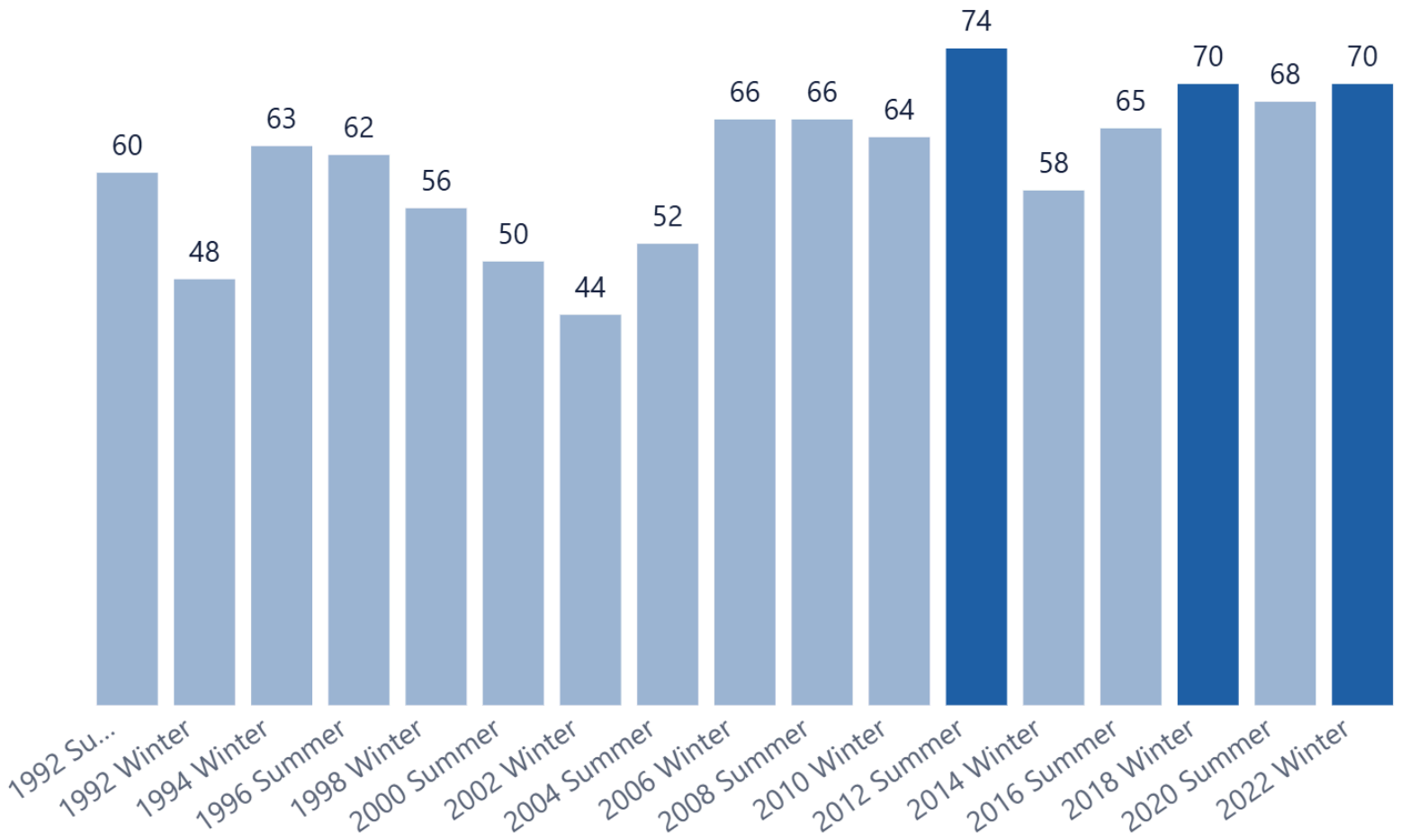


Only one nation (Ethiopia) is low income, so read the Low bar with care.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 8

Medals per Games rose from 54 to 67 after 2004

Medals by Games



54.4
medals per Games, 1992-2004

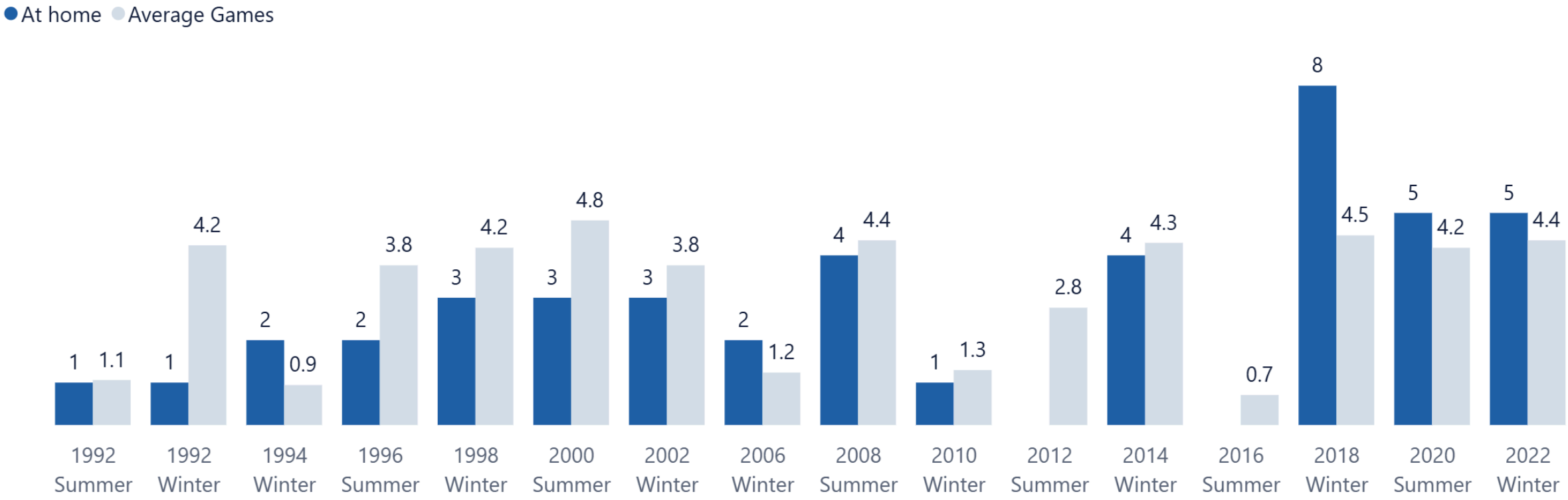
66.8
medals per Games, 2006-2022

Entries rose only 8%.
From 1,512 to 1,636 per Games while medals rose 23%, so each entry became more likely to win.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 9

Hosts win no more at home: 2.75 medals per home Games against 3.15 on average

Host nation medals at its home Games vs its average Games



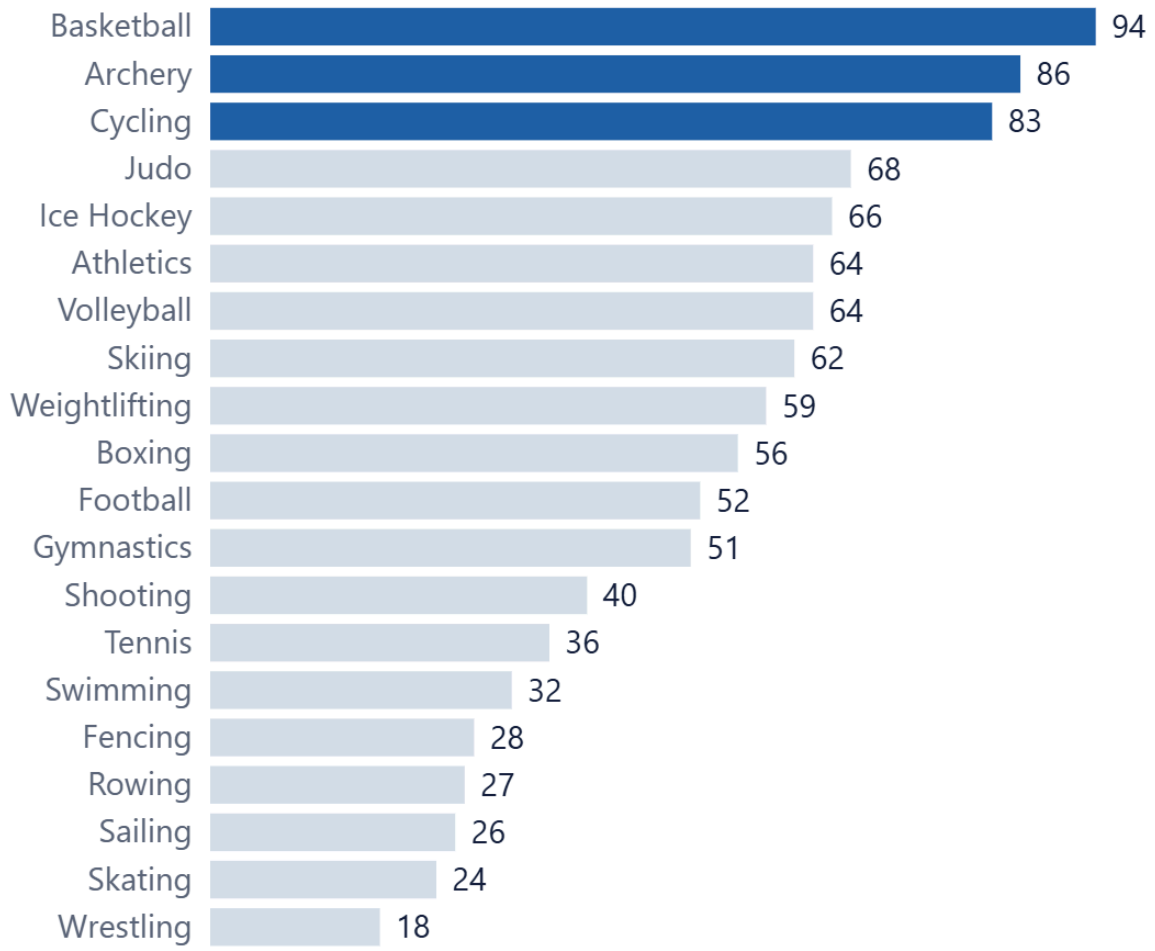
11 of 16 hosts won fewer medals at home than in an average Games; Korea 2018 (8 against 4.5) is the clear exception. Greece (2004) is not in the nation list.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 10

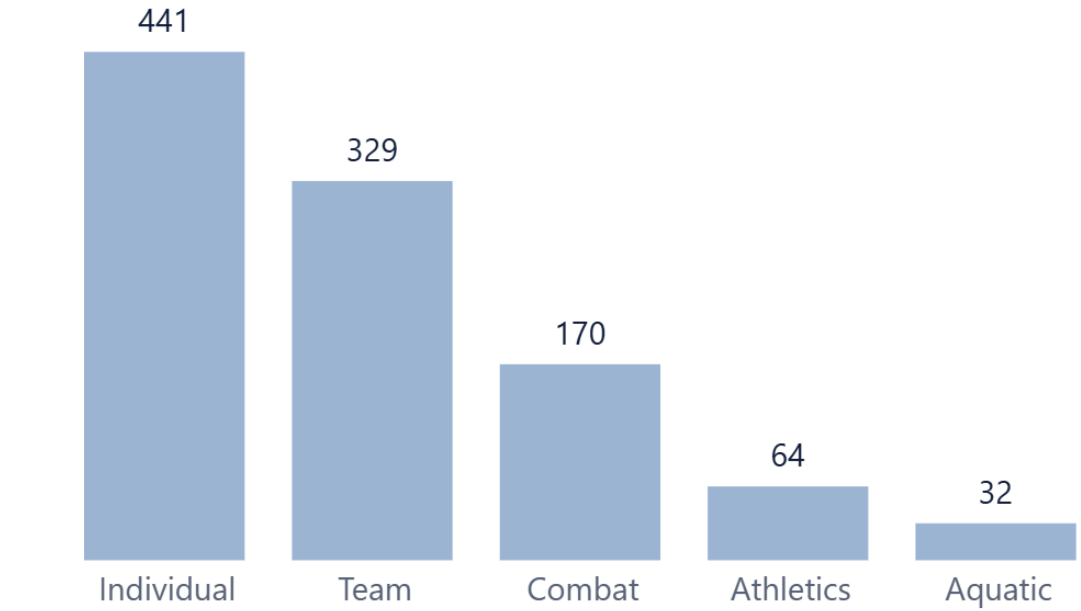
Dashboards

Medals follow the number of events: basketball, archery and cycling lead

Medals by sport



Medals by sport family



Basketball holds 12 events, archery 10, cycling 11; wrestling holds 4.

Count events before judging a sport as a medal source.

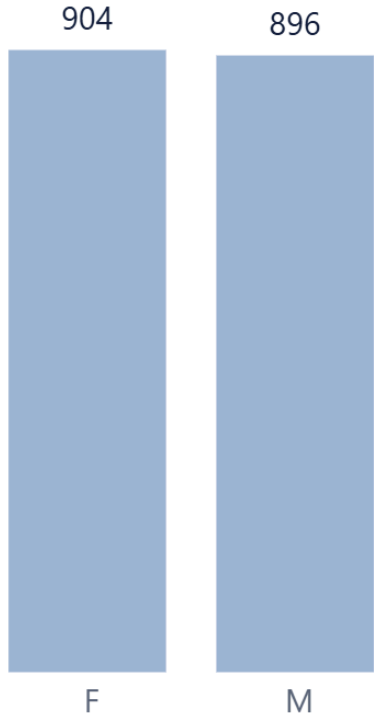
Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 11

Women and men win medals almost equally: 492 against 546

Medals by sex



Athletes by sex



Medals per 100 entries



Age is not analysed: birth years do not match the Games (ages run from -13 to 57), and 78 athletes have weight 0.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 1,800 athletes · 12

Dashboards

Three actions: fund conversion, benchmark per head, fix the data

1

Fund conversion, not entries

Every nation enters about 900 times; the leaders win 6-9% of those entries. Put money where athletes already reach finals.

Target: lift mid-table conversion from 2% to 4%

2

Benchmark per head

Judge smaller nations on medals per million people and per $bn GDP, where Jamaica, New Zealand and Kenya lead.

Target: per-capita goals, not raw medal counts

3

Fix the data first

Only 13% of event-Games have medals and birth years do not match the Games. Reload a complete list before forecasting.

Target: medals for all 2,363 event-Games

3.9%
conversion today, all nations

0.21
medals per million people

60.9
medals per Games

Source: Olympics model (Project_06_Olympic_Games.xlsx) · targets are planning goals, not forecasts · 13

Assumptions and data issues behind these numbers

Corrections to the original report

- Total Medals counted every entry (23K); it now counts medals only (1,036).
- Medals per million and per $bn GDP divided by a million and a billion twice; fixed.
- The old pages said the USA leads; on true medals Australia leads and the USA is 8th.
- Forecast pages are dropped: the data cannot support a forecast.

Data limits

- Only 315 of 2,363 event-Games have medals, and 67 award the same medal twice.
- Every sport appears in Summer and Winter, so season splits mean little.
- Ages run from -13 to 57 and 78 athletes weigh 0 kg; body and age measures are not used.
- The source workbook is missing, so the data cannot be refreshed.

Summer and Winter side by side

Season	Games	Entries	Medals	Per Games
Summer	8	12,616	497	62.1
Winter	9	14,208	539	59.9

Treat this as a teaching dataset.

The patterns compare nations inside this data. They are not a record of real Olympic history: real Games award a gold, silver and bronze in every event, and sports belong to one season.

What still holds

Comparisons between nations, regions and income groups, because every nation is measured the same way.

What to do next

Restore the source workbook and reload a complete medal list before using the report for decisions.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · all entries · 14

MEDALS · 30 NATIONS · 17 GAMES · 1992-2022

Nine nations win most medals

Season: All | Games: All | Region: All

1,036 medals won

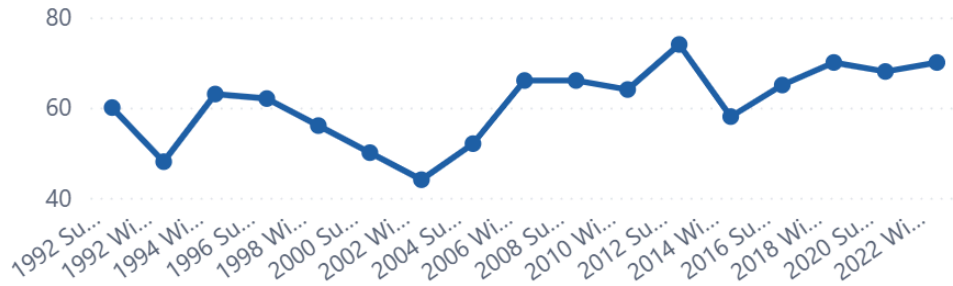
334 gold medals

26,824 entries

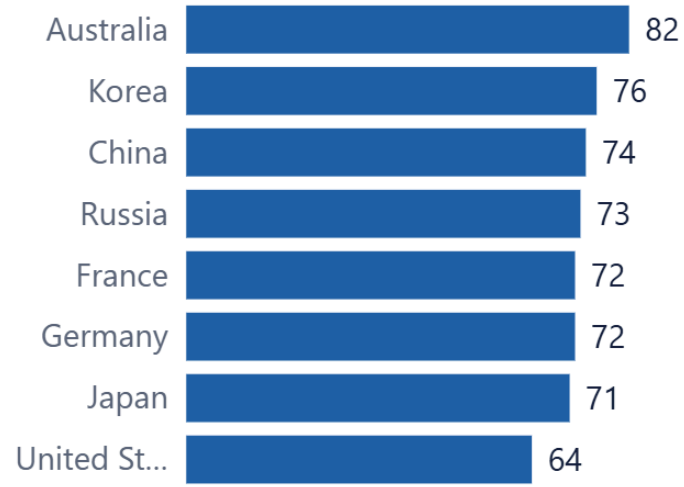
3.9% of entries win a medal

30 nations

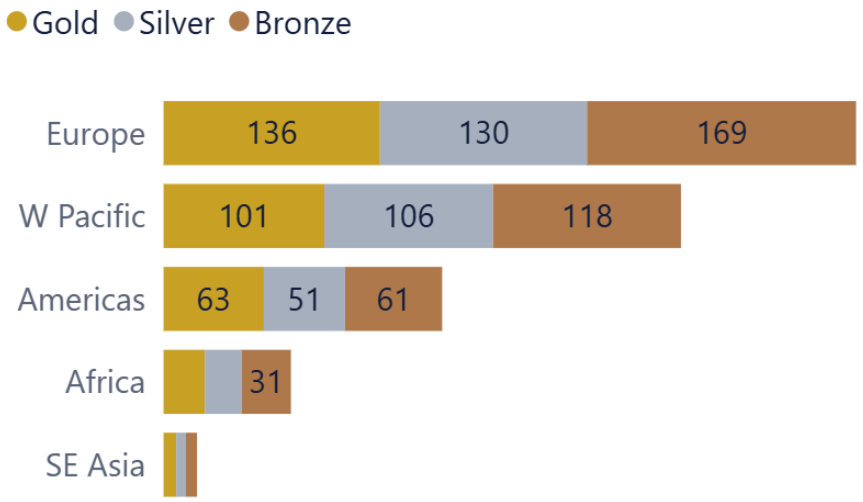
Medals by Games



Top 8 nations (right-click for profile)



Medal mix by region



WHAT THIS SHOWS

1,036 medals from 26,824 entries: 3.9% win.

Nine nations win 61% of medals and 62% of golds.

Australia leads with 82 medals.

Story | **Medals** | Nations | Sports

Taiwo Ajayi
Report author



NATIONS · 30 NATIONS · 17 GAMES · 1992-2022

Conversion separates the leaders

Season: All | Games: All | Region: All

Australia

nation with the most medals

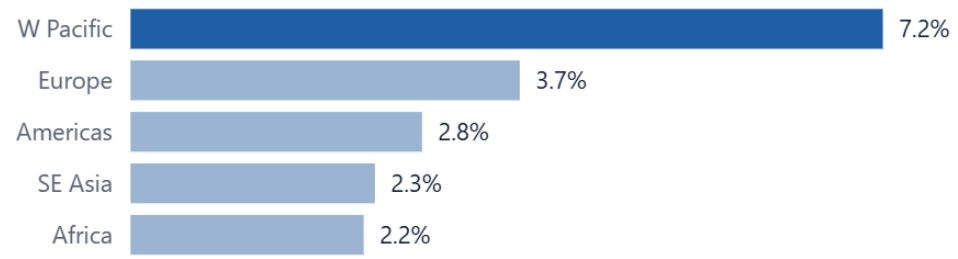
60.9 medals per Games

32.2% of medals are gold

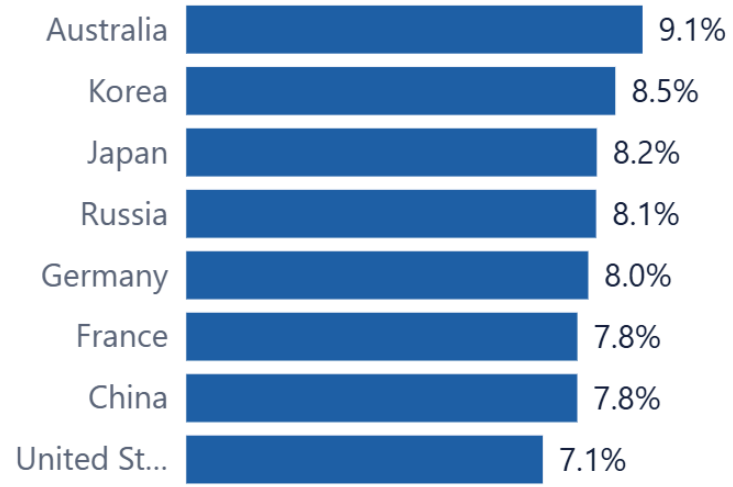
0.21 per million people

0.012 medals per $bn GDP

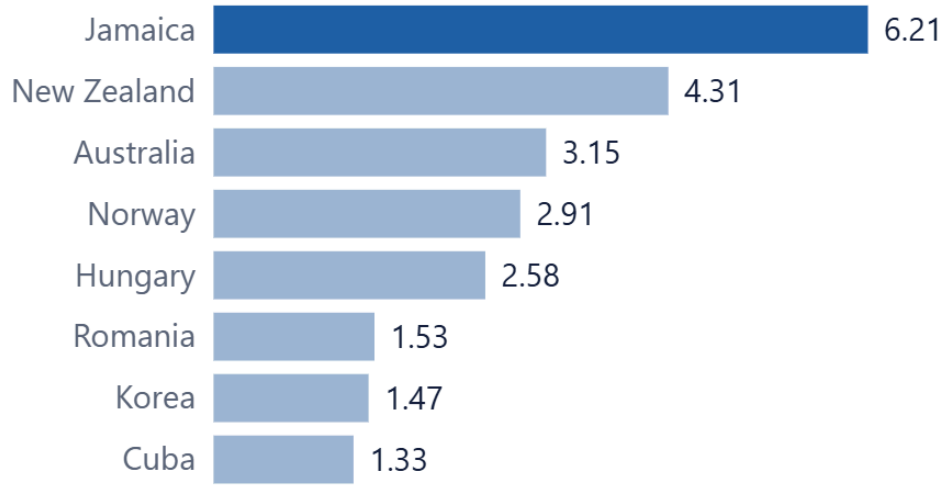
Medals per 100 entries, by region



Conversion, top 8 (right-click for profile)



Medals per million people, top 8



WHAT THIS SHOWS

Leaders convert 6-9% of entries; the rest 1-3%.

Jamaica wins 6.2 medals per million people.

W Pacific converts best (7.2%), Africa least (2.2%).

Story | Medals | Nations | Sports

Taiwo Ajayi
Report author



SPORTS · 30 NATIONS · 17 GAMES · 1992-2022

Medals follow the events

Season: All | Games: All | Region: All

767
athletes with a medal

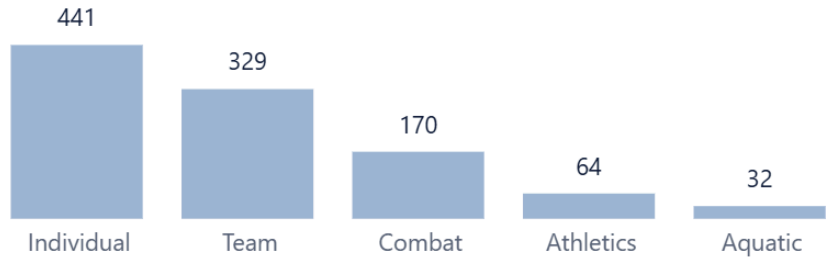
1,800
athletes competing

32.2%
of medals are gold

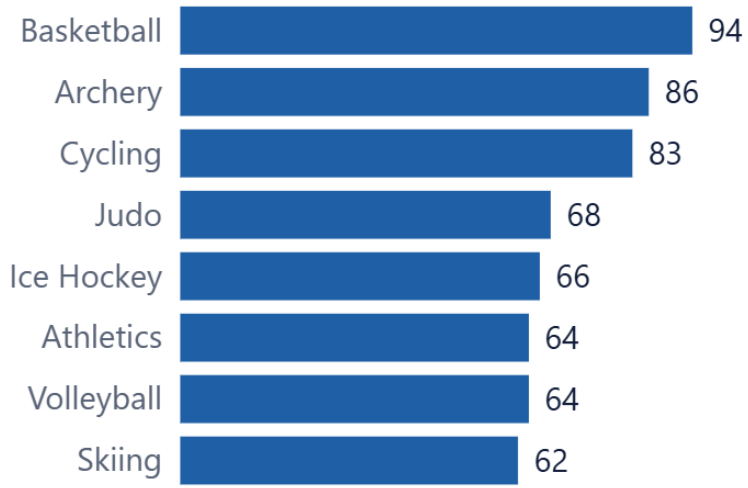
25.4%
from top 3 sports

3.9%
of entries win a medal

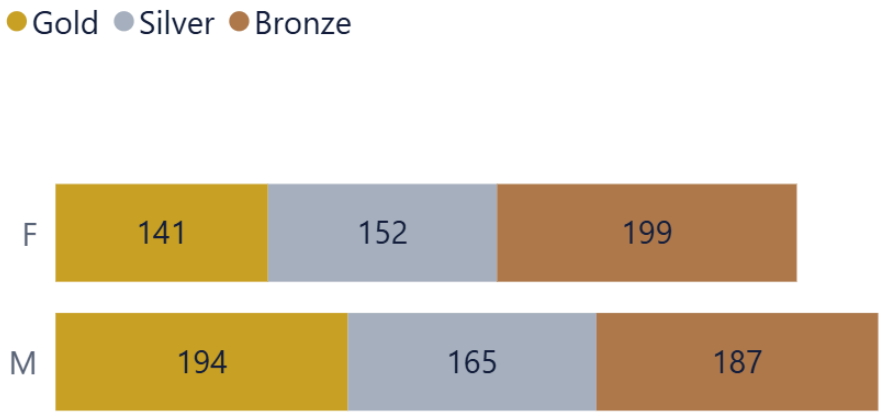
Medals by sport family



Top 8 sports by medals



Medal mix by sex



WHAT THIS SHOWS

Basketball, archery and cycling lead because they hold the most events.

Women won 492 medals, men 546.

A quarter of medals come from the top three sports.

Story | Medals | Nations | Sports

Taiwo Ajayi
Report author

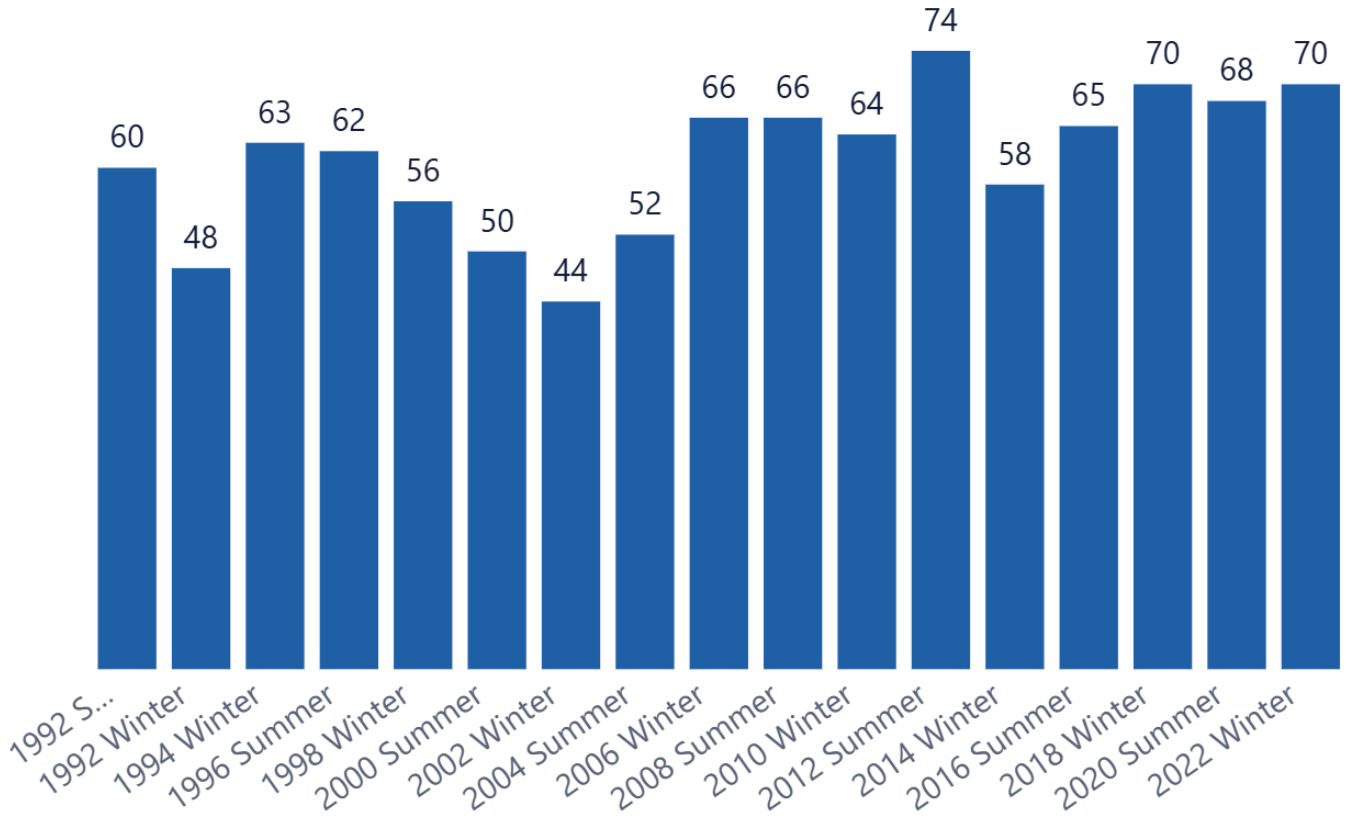


Nation profile: medals, conversion and sports for the selected nation

Back

1,036	334	3.9%	60.9	0.21
medals	gold	of entries win a medal	medals per Games	per million people

Medals by Games



Medals by sport

